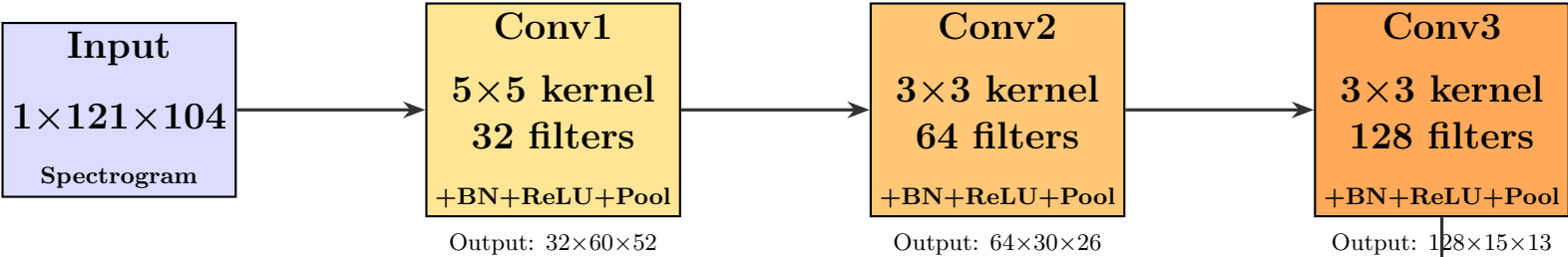


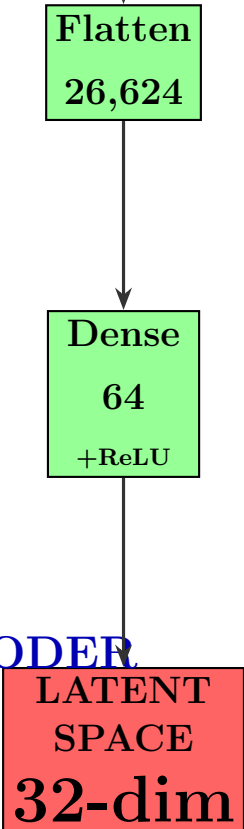
Bowhead Whale Autoencoder Architecture

Hybrid 5×5/3×3 Convolutional Autoencoder

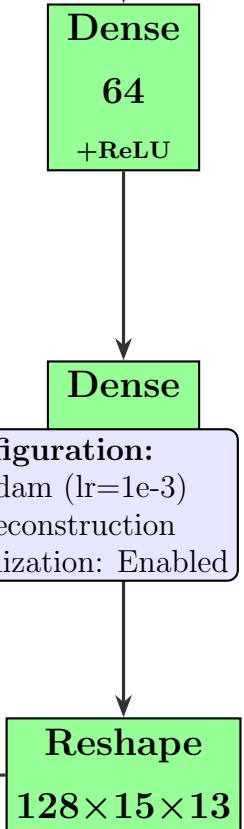
ENCODER



BOTTLENECK



DECODER



Hybrid Architecture:
First layer uses 5×5 kernel to capture broad N/U-shaped whale call patterns
Subsequent layers use 3×3 kernels for efficient detail refinement

Model Statistics:

- Parameters: 3,359,105
- Latent Dimension: 32
- Compression Ratio: 378:1

Training Configuration:

- Optimizer: Adam (lr=1e-3)
- Loss: MSE Reconstruction
- Batch Normalization: Enabled

